University of Michigan Ross · Admissions artifact

Hi, Admissions Officer. Here is my artifact.

I’m Ayaan Ahmed. This is Project Water.

I taught myself 3D design, animated a proposed solar-powered pumping and filtration system, and earned unanimous approval from SolarPak’s six-member council.

View the live artifact

An independent idea. A shared commitment.

PROJECT WATER.

Could solar energy help a rural community access filtered water more reliably?

As CEO of SolarPak, I oversee our installation efforts. Project Water is my proposal to use solar energy to pump, filter and store water for a rural community.

An approved concept. Implementation is still planned.

The origin story

It started with a question.

Through my work with SolarPak, I saw how solar energy could help homes facing unreliable electricity. Access to water was another challenge in the communities we served.

What if the same energy could help bring filtered water to a community?

I began exploring a system that could use solar energy to pump water through a filter into a raised tank, then use gravity to carry it to taps.







Context photographs · Pakistan · Sources and licenses at the end of this PDF

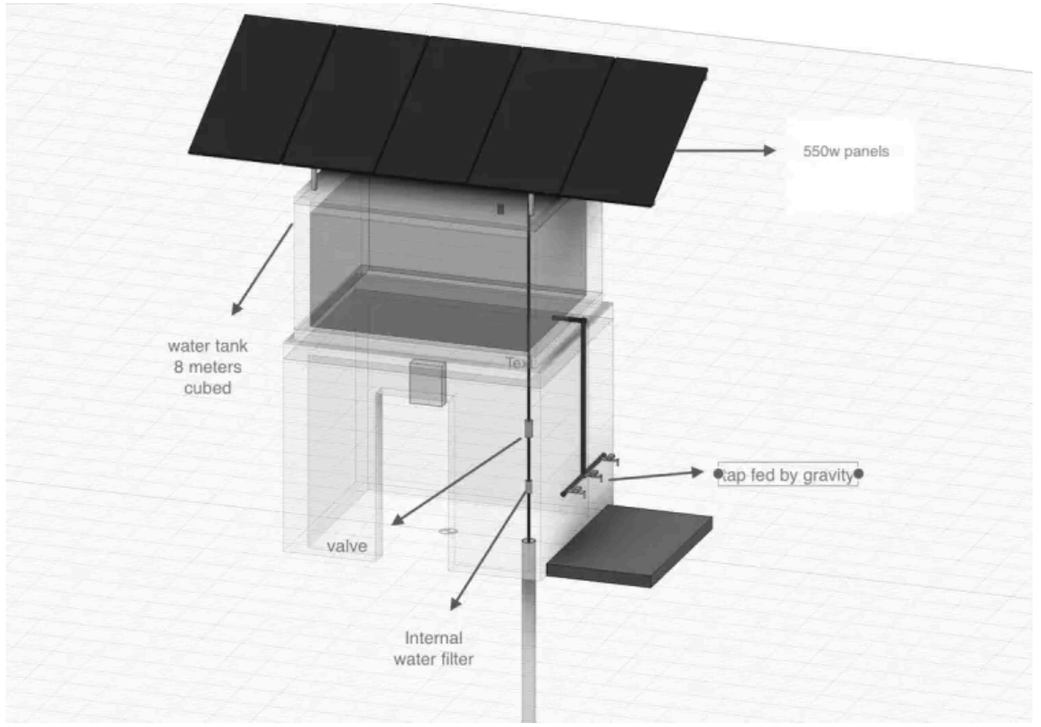
The design

I learned by making.

The idea is simple: use sunlight to pump and filter water, store it in a raised tank, then let gravity carry it to taps.

I had never designed in 3D before.

Over two weeks, I taught myself Autodesk Fusion 360, a tool for designing objects in 3D, and developed my original water-system concept.



My original 3D design · includes an internal water filter

The decision

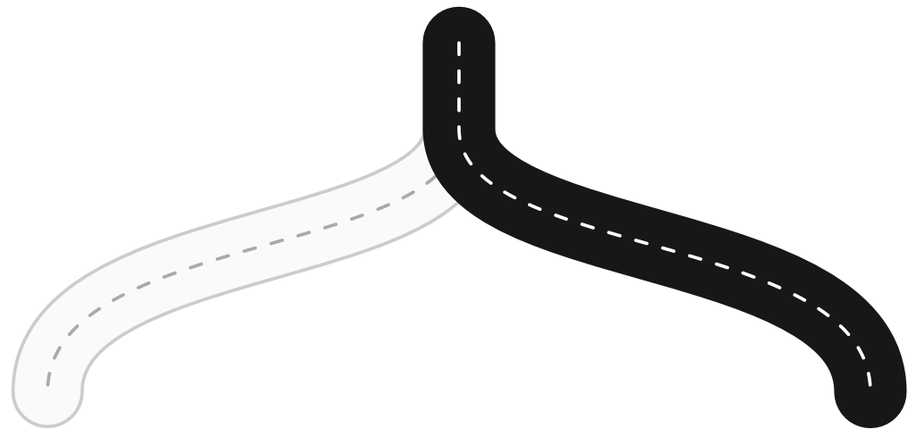
Choosing where to invest.

I had to decide whether to keep funding installations for individual households or invest in developing a shared water system.

I chose to explore Project Water because it could reach a whole village of around 200 people. That potential made it worth pursuing to me, even though it still needs to be built and tested.

I made a 3D design and an animation to explain the pumping and filtration proposal to SolarPak’s council and ask for their support.

What would you choose?



01

Fund installations

Support individual households through our existing work.

02

Develop a new system

Explore solar-powered pumping and filtration for a village.

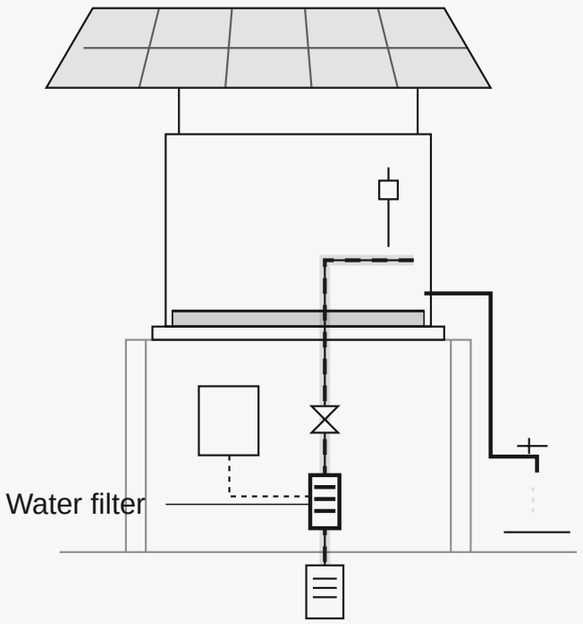
That’s the path I chose.

The possibility of reaching a whole village made it worth exploring to me. The system is intended for around 200 people. I’m still waiting to see if that risk will pay off.

See the system I developed ↘

The idea, in four steps.

A 14-second overview. Choose a step to explore.



An illustration of the proposed pumping, filtration, storage and delivery system. Implementation is still planned.

Watch the animation on the website

Use sunlight | **Filter water**

Store water | Reach the taps

Filter the water.

Water passes through a filter before it enters the storage tank. Filtration is a core part of the proposed system, alongside reliable access to water.

Earning support

From concept to commitment.

I presented my design and animation to SolarPak’s six-member council. All six approved the proposed solar-powered pumping and filtration system.

6 / 6	PKR 400,000	200
Council approval	Approximately raised by SolarPak overall	Intended residents · not yet served

A target village has been identified. Implementation is tentatively planned for mid-November 2026, subject to technical verification and contractor arrangements.

What I learned

I thought this was worth exploring because it could help a whole village.

I’m still waiting to see if that risk will pay off.

I learned that making the design and getting people to back it are different challenges. My next responsibility is to oversee the engineers as we work toward building the pumping and filtration system.

Ayaan Ahmed

Photograph credits.

People returning home as soon as the water recedes enough

Department for International Development / Magnus Wolfe-Murray · 2010

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Hanna lake and water Pollution

Izharkasi · 2015

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Water pond in Village Kertee

Gumansingh12340 · 2017

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Displayed in grayscale. These photographs provide context; they do not document Project Water.

View the live artifact

PLANNED FOR
MID NOVEMBER
2026

Tentative implementation target